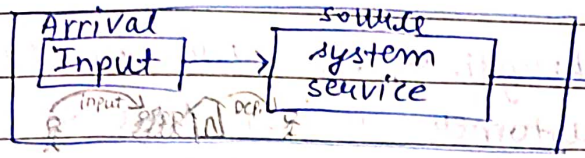
Exponential Dist. $E(X) = \lambda e^{-\lambda x}$

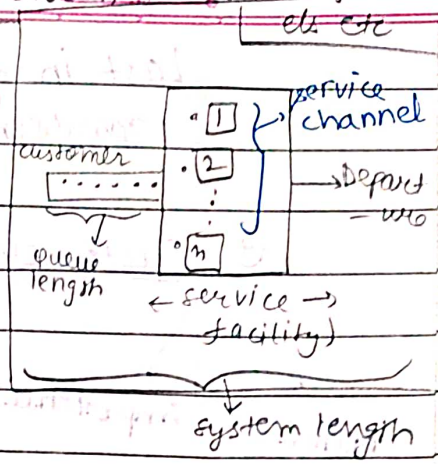
queue → customers waiting to be serviced. This does not include the customer being serviced.

customer → The arriving unit that requires some services to be performed is called customer. It may be persons, machines, vech etc etc

→ Queuing theory

It is the study of behaviour of queue





Queuing system

→ The queuing system have the four components ~~in~~

① input i.e arrival pattern of customer

→ The input is described by the way in which the customer arrive and join the system. generally customers arrive in random manner which is not possible to predict. Thus, arrival pattern can be described in terms of probabilities and consequently the probability distribution for inter arrival time (the time b/w two successive arrivals)

In queuing theory we assuming that customers arriving in poission processes with the mean arrival rate ~~in the~~ denoted by λ

mean arrival rate $= \lambda$ → no. of customer arriving per unit time.

② service mechanism (pattern) = It means the arrangement of service facilities to serve customers if there are infinite no. of servers than all the customers will be serve instantaneously which is not feasible. If the number is finite. then the customer will be serve ~~in some specific order~~ in some specific order with service time a constant or random variable. Distribution of service time follows exponential distribution with mean services rate $= \frac{1}{\lambda}$

single channel, multiple channel

③ Queuing diciplane :- It is the set of rules acc. to which customers are selected for service when a queue has been form the most common diciplane are 'first come first serve' (FCFS) (FIFO), first in first out (FIFO)

Waiting time :- The average time for which a customer has to wait in the queue to get service.

System time :- service time + waiting time

Last in first out (LIFO), selection for service in random order (SIRO),

④ Customer behaviour :- Length of the queue or the waiting time of the customer or ideal time of service facility dependens upon the behaviour of the customer. The behaviour refers to the impatience of the customer during a stay in line. Customer behaviour can be defined as :-

Patience | Impatience

① Balking - If the queue length is very large to a customer he/she may not join the queue this is known as the balking

② Reneging - Sometime a customer who is already in the queue will leave the queue if antisipation of longer waiting line this kind of departure is known as Reneging

③ Jockeying - If there is more than one queue the customer may switch to one to another queue in order to get serve erlier. perhaps the situation may be that queue which he join later is souter but the service is very slow and again switch to the another queue this behaviour of customer is known as jockeying.

④ Bulk arrival or collusion = when a customer arriving group and

Mean service rate (μ) → No of customer served per unit of time

colleborate with each other that only any one of them will stand in the queue is known as bulk arrival or collusion.

- Arrivals
  - Nature of arrivals
    - Size of arrivals
      - Finite population
      - Infinite Population
    - Arrival time distribution
      - Constant (Deterministic)
      - Random Probablistic
  - Capacity of system
    - limited
    - unlimited
  - customer behaviour
    - Balking
    - reneging
    - Jokeying
    - collus-ion

- service mechanism
  - Service facility design
    - single channel
    - multi channel
  - Queue disciplane
    - FIFO
    - LIFO
    - SIRO
    - collusio-n

behaviour time dependent

behaviour time independent

~~Transiant~~ State & Steady state

Transient

the distribution of customers arrival time and service are two constituents consisting of steady of waiting line. Under the fixed condition of customer arrival and service facility a queue length is a function of line. Such a queue system can be considered for some random experiments and the various events of the experi-ment can be taken to various changes occuring in the

system at any time. And we can identify the transian state and the steady state in the system.

Transian state in waiting line
queuing theory analysis involves the study of a system behaviour ~~of~~ overtime a system is said to be in transion state when its operating characteristics on behaviour depends on time. this happens usually at the initial stage of operation of the system where its behaviour still depends upon the initial conditions

$(\lambda < \mu)$ steady state = The system will settle down as steady state when the rate of arrivals of customers is less than the rate of service and both are constant. The system become independent of initial state of the queue than the probability of finding a perticular length of the queue at any time will be

→ The necessary condition for steady state to be reached is that elape time since the start of the operations become sufficiently large i.e $t \to \infty$ but this condition is not sufficient as the existance of steady state also depends upon the behaviour of the system. If the rate of arrival is greater than the rate of service than a steady state cannot be reached.
Hence, we assume that a system ~~req~~ acquires study state as $t \to \infty$ and when the no. of arrivals during certain time interval become independent of t

i.e $\lim_{t \to \infty} P_n(t) \longrightarrow P_n$

$P_n \to$ Probability of n customers waiting for the service